

Al-Nahrain University
College of Engineering
Department of Computer
Fourth Stage/First Semester



Control Engineering I
Experiment No.3: Stability and the Effect of Loop Gain

Course Lecturer : Dr. Mushtaq Talib
Laboratory Assistant:Ghaith Ali
Name of student:Aisha Ammar Abdalrhman
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Experiment No. 3

Stability and the Effect of Loop Gain

Objectives:

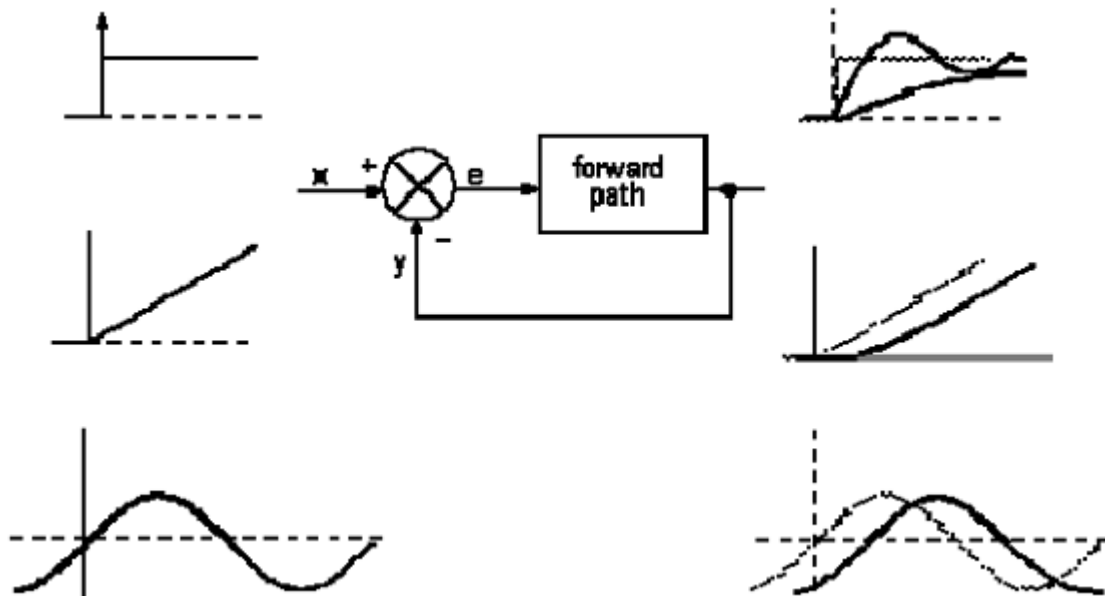
To learn the measurement of transient response within a closed loop control system for step inputs. To learn the effects of loop gain on system's stability

Procedure:

The Control & Instrumentation Principles 33-033 application was opened on the computer, and the option *Stability and the Effect of Loop Gain* was selected.

Practical 1: Measuring Transient Response

A general system was shown, where an input x and an output y were compared to give an error (e) with the relation [$e = x - y$].”

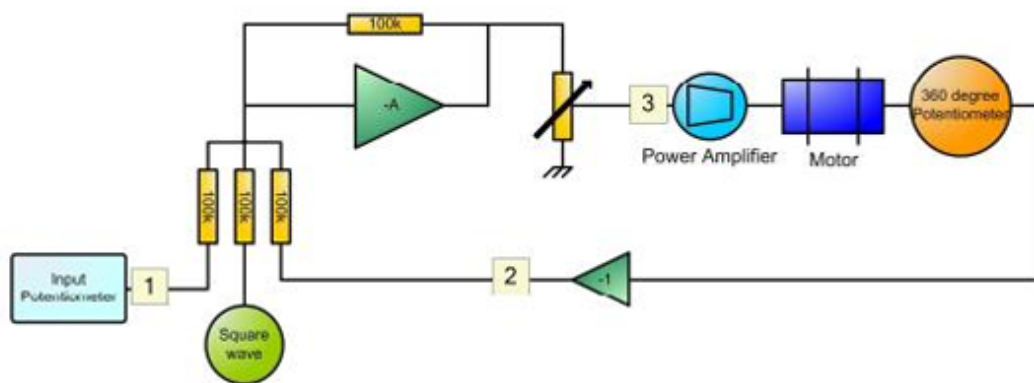


The process of comparison was represented by conventional symbols as in the diagram, where the input and output were not necessarily required to be voltage signals. For a purely electrical system, the comparator was considered to be an operational amplifier.

The error was operated on in the forward path, which was defined as everything between the error and the final output y . Thus, the forward path was described as containing a facility by which the error was converted to a voltage, followed by a gain stage and a power amplifier by which a motor was driven, and then some reduction gear by which the output shaft was operated.

The design and performance characteristics of systems were often considered in terms of the response to a step or ramp input, with possible responses shown above, where the input was shown in shadow.

The connections of the block diagram were made, and the instructions of Practical 1 were followed.



Practical 2: The effect of Gain on Stability

The step response of a system was considered to provide useful information about the general system characteristics and was often required to meet certain requirements in the system performance specification. To generate the step response, a very simple input was required, and for these reasons much consideration was given to the step response.

For a given system, the form of the step response was greatly affected by the system gain. The gain was understood to determine how much power was applied to move the output for a given error.

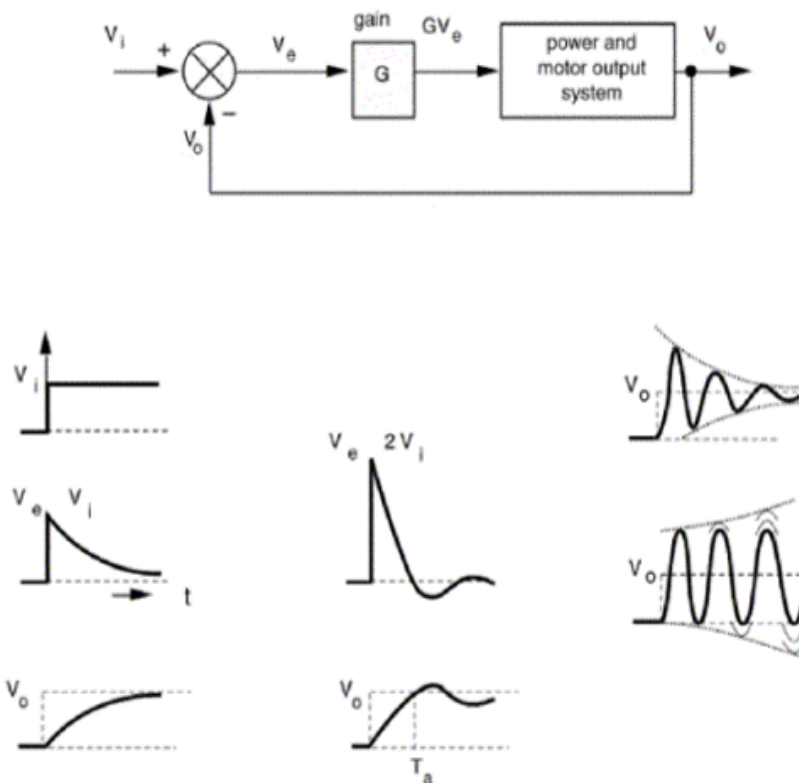
A purely electrical position control system was represented as shown below, where the input (V_i), output (V_o), and error (V_e) were all shown as voltages, and the forward path gain (G) was shown separately, with the voltage applied to the power amplifier being ($G V_e$).

If the step input (V_i) was applied, the initial value of the error was made equal to (V_i), since (V_o) was zero. If the gain was 1, the power amplifier input was initially V_i , and as the motor was rotated the output was gradually aligned with the input, with the motor being slowed as the error was decreased.

If the gain was 2, the initial input to the power amplifier was ($2V_i$), causing the motor to be moved faster, and although the error was decreased, the motor was likely to overshoot the required final position due to the delay in the motor.

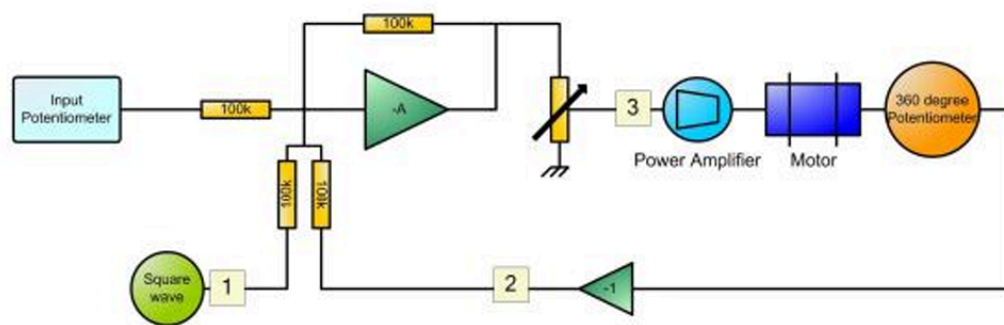
When the motor was finally stopped, the error was reversed in sense, so that the motor was reversed and the system was aligned or was made to undershoot, but was eventually settled.

If the gain was increased further, several oscillations were required for the system to be settled. If two delays were present in the system, a steady oscillation at the output was produced, or even an increasing oscillation was created, and such systems were considered useless for control purposes.



A closed loop control system response to a step input is sluggish if the gain is low. An increase in the gain causes a faster response, but if too much gain is applied the response may overshoot and take several oscillations before settling. Further increase in gain may cause sustained oscillations to build up.

The connections of the following block diagram were made, and the instructions of Practical 2 were followed.



Discussion:

Q1/ Explain in details the effect of adding gain in control system?

Adding gain in a control system strongly affects the system's stability and dynamic behavior. When you increase the gain, the controller applies more corrective action to the system error, which makes the output respond faster. As a result, the system reaches the desired value more quickly.

However, increasing the gain too much can cause overshoot because the system moves too forcefully before it can slow down. Further increases in gain can create oscillations, as the system overcorrects and does not have enough time to settle smoothly. In some cases, these oscillations are damped and the system eventually settles, but in other cases, the oscillations persist or grow, leading to instability.

Therefore, increasing gain improves the speed of response, but excessive gain reduces stability and can cause overshoot, oscillations, or complete loss of control.